Theory of Computation

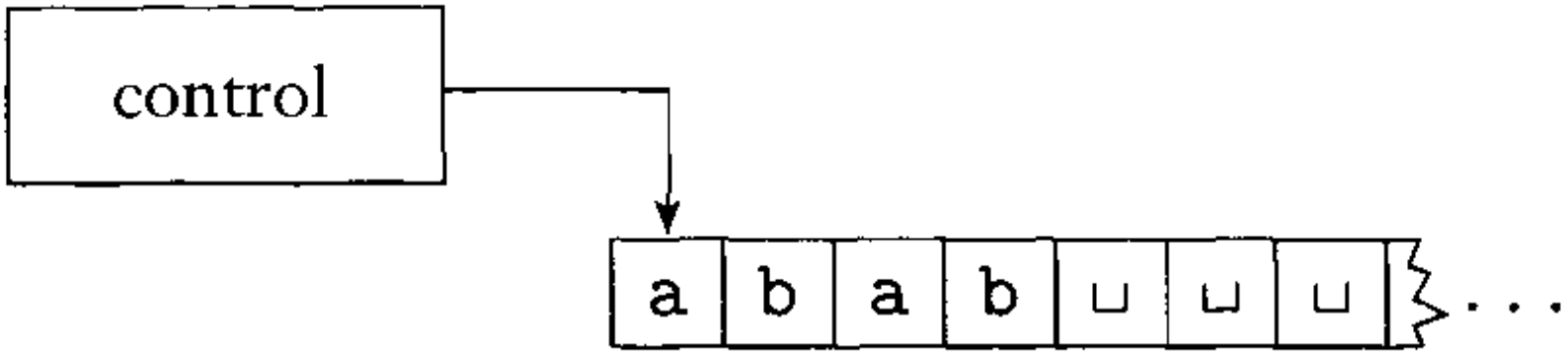
What can be done by computing machines?
Computability

What can realistically be done by computing machines?
Complexity

Where on earth to start?
Turing Machines

Edwin Blake, Tommie Meyer, Nathan Geffen & Sonia Berman

A Turing Machine

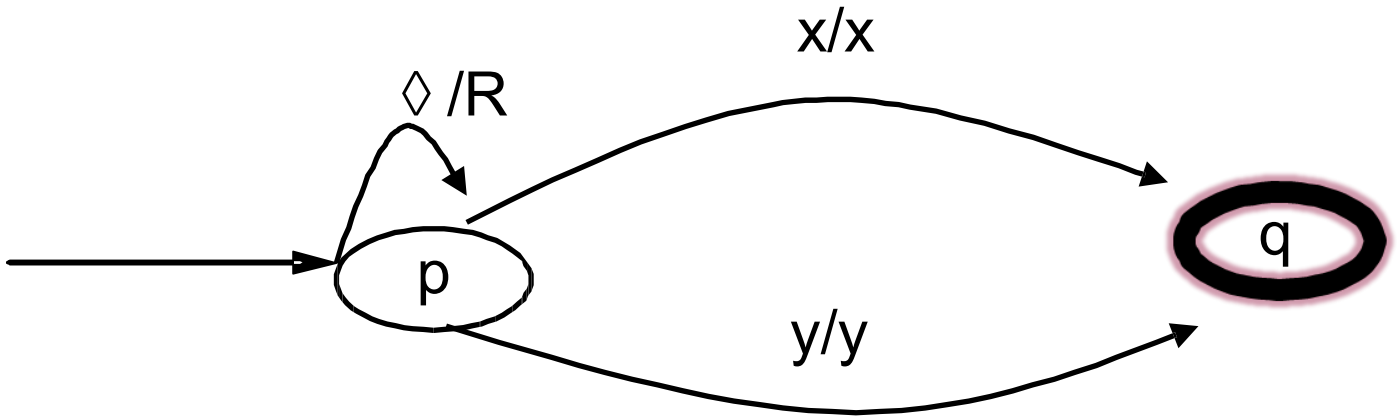


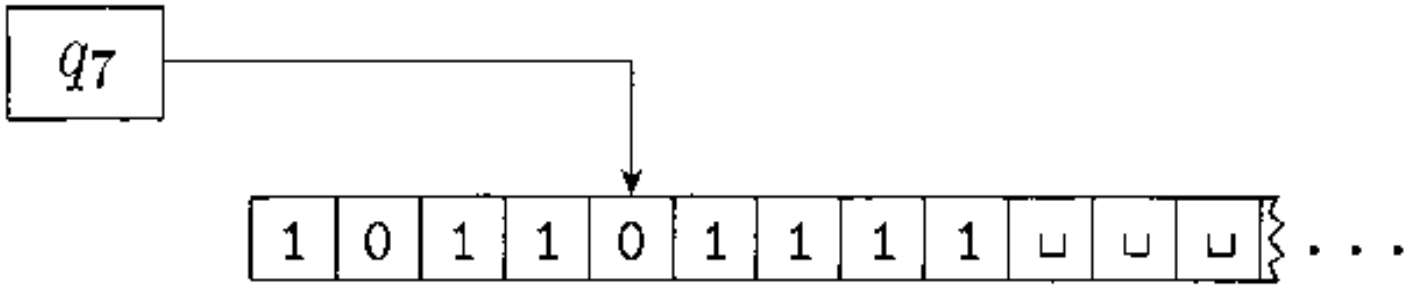
$\delta(p, \Diamond) = (p, R)$

$\delta(p, x) = (q, x)$

$\delta(p, y) = (q, y)$

Current	Read	Action	Next
p	x	x	q
p	y	y	q
p	◊	R	p





A Turing machine with configuration $1011q_701111$

The configuration shows both the state and the tape head position by putting the state name just before the current cell

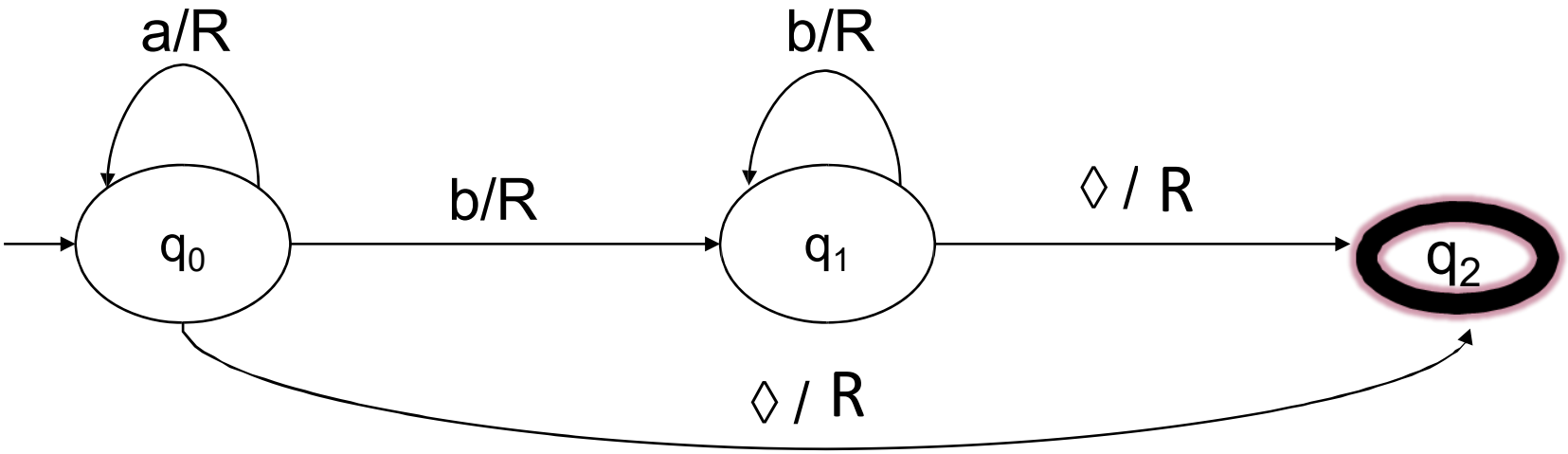
Suppose we had $\delta(q_7, 0) = (q_2, R)$

If the current configuration is $1011\boldsymbol{q_7}\mathbf{0}1111$ what will the next configuration be?

$10110\boldsymbol{q_2}\mathbf{1}1111$

Example of a Turing Machine

- Input symbols = {a,b}
- Alphabet = {◊, a, b} (◊ = blank)
- Missing transition → implies crash
- Can you see what language this machine accepts?



Accepts the language denoted by the regular expression a*b*

Regular Expressions

Consider the English alphabet = { “a”, “b”, “c”, … “z” }

Alternation:

a | b | c | d | … | z = { “a”, “b”, “c”, …, “z” }

Concatenation:

d • o • g = “dog”

(d • o • g) | (c • a • t) = {“dog”, “cat”}

Concatenation symbol may be omitted

Repetition:

a* means 0 or more a (**Kleene Closure**)

a* = {“”, “a”, “aa”, “aaa”, … }

String of all a’s with at least one a: a • a* = { “a”, “aa”, “aaa”, …}

Empty string: epsilon symbol (ε)

ε = “”

a | ε represents {“”, “a”}

(a • b) | ε represents {“”, “ab”}

Regular Expressions

Abbreviations

[abcd] means (a | b | c | d)

[b-g] means [bcdefg]

[b-gM-Qkr] means [bcdefgMNOPQkr]

M? means (M | ε), can be read as “optionally M”

M+ means (M • M*), can be read as “one or more M”

. the period symbol means any single character except newline

Do these add power to Regular Expressions?

No as they build on already defined operations

Just convenient

Regex: Operator Precedence

Operation	Description	Precedence order
r^* , r^+	closures	highest
r s	concatenation	medium
r | s	alternation	lowest

Examples (input alphabet = {a, b})

What's denoted by the following regular expressions?

1. **(a|b)(a|b)**
2. **(a|b)***
3. **(ab)***
4. **a|a*b**

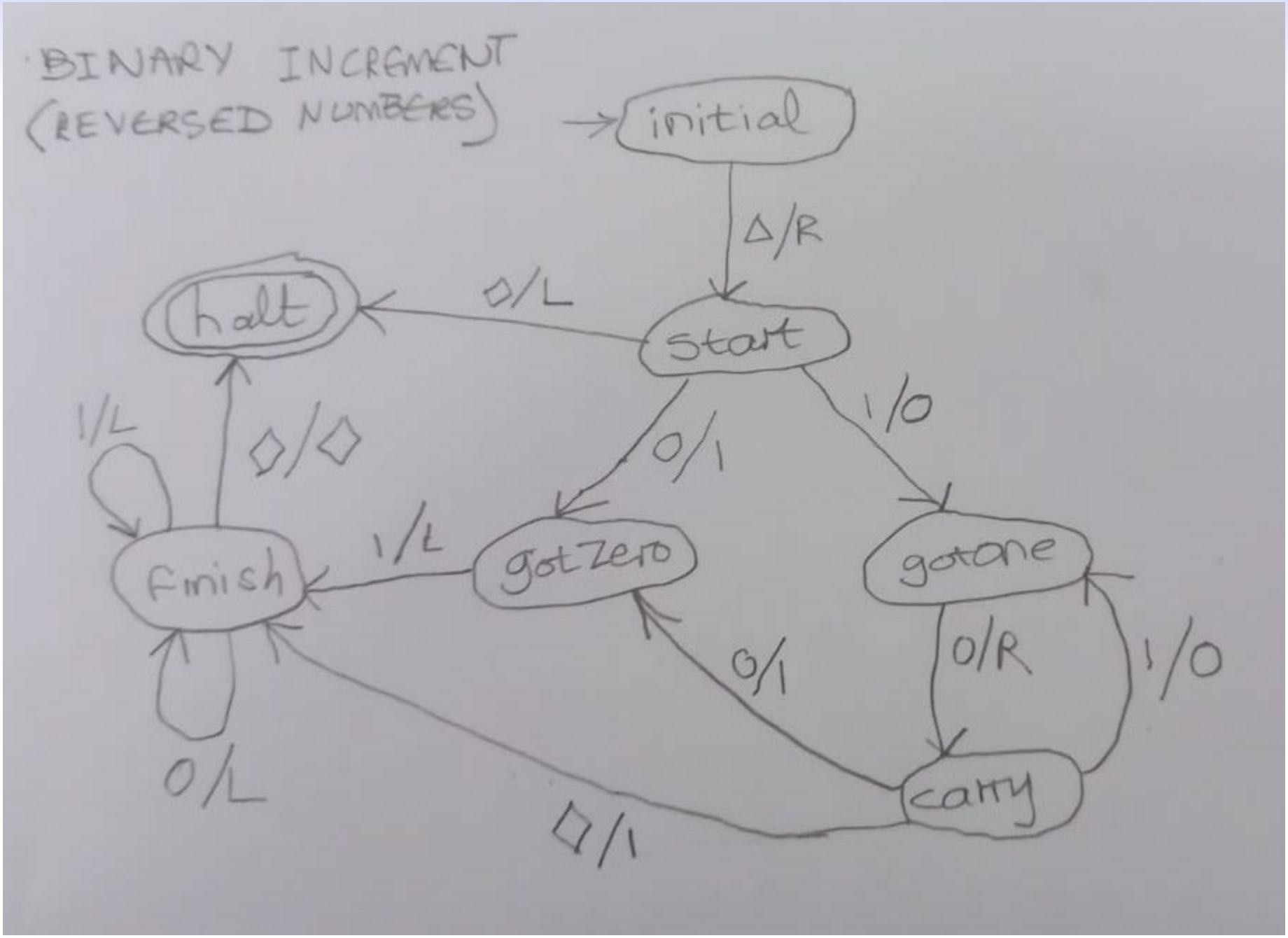
Examples (input alphabet = {a, b})

What's denoted by the following regular expressions?

1. **(a|b)(a|b)**
2. **(a|b)***
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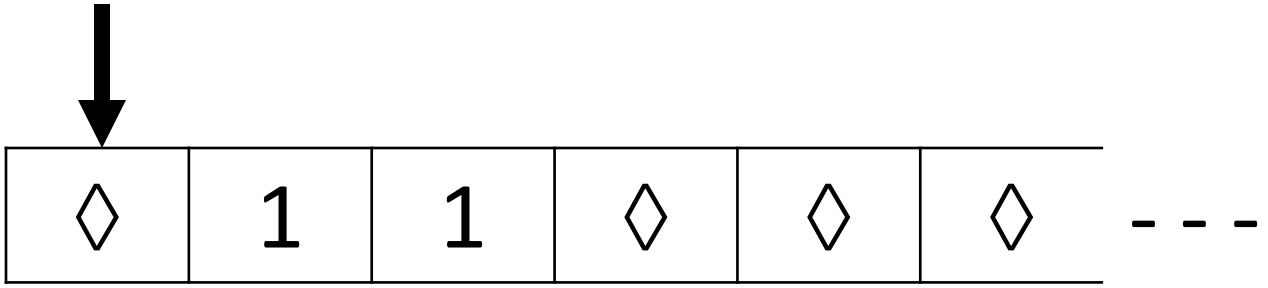
1. **aa or ab or ba or bb**
2. **A string of any length comprising only the characters a or b in any arrangement**
3. **Empty string, or ab, or abab, or abababab …**
4. **Either a single a, or a single b preceded by any number of a characters**

No.	Binary	Reversed	Incremented	
0	0	◊0◊	◊1◊	Flip until first 0 flipped
1	1	◊1◊	◊01◊	If flip and no 0, change ◊ to 1
2	10	◊01◊	◊11◊	Flip until first 0 flipped
3	11	◊11◊	◊001◊	If flip and no 0, change ◊ to 1
4	100	◊001◊	◊101◊	Flip until first 0 flipped
5	101	◊101◊	◊011◊	Flip until first 0 flipped
6	110	◊011◊	◊111◊	Flip until first 0 flipped
7	111	◊111◊	◊0001◊	If flip and no 0, change ◊ to 1
8	1000	◊0001◊	◊1001◊	Flip until first 0 flipped
9	1001	◊1001◊	◊0101◊	Flip until first 0 flipped
10	1010	◊0101◊	◊1101◊	Flip until first 0 flipped
11	1011	◊1101◊	◊0011◊	Flip until first 0 flipped
12	1100	◊0011◊	◊1011◊	Flip until first 0 flipped
13	1101	◊1011◊	◊0111◊	Flip until first 0 flipped

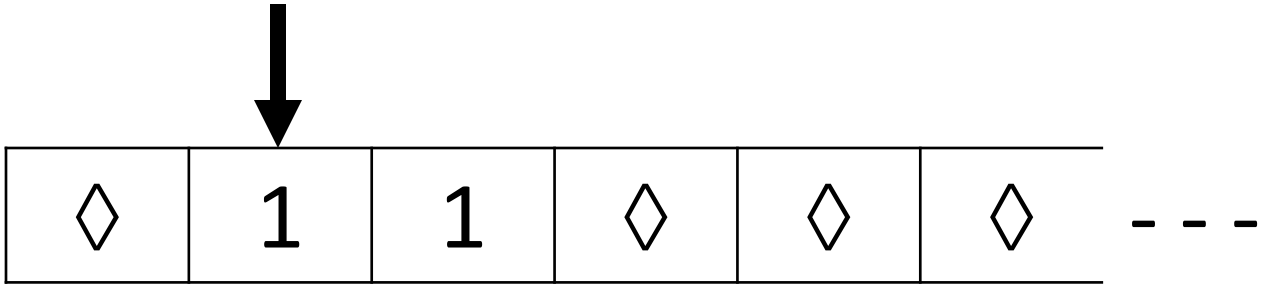

BINARY INCREMENT
(REVERSED NUMBERS)
initial
Δ/R
halt
□/L
start
1/L
□/□
0/1
1/0
finish
1/L
gotZero
gotone
0/L
0/1
0/R
1/0
carry
□/1

Incrementing a *Reversed* Binary Value

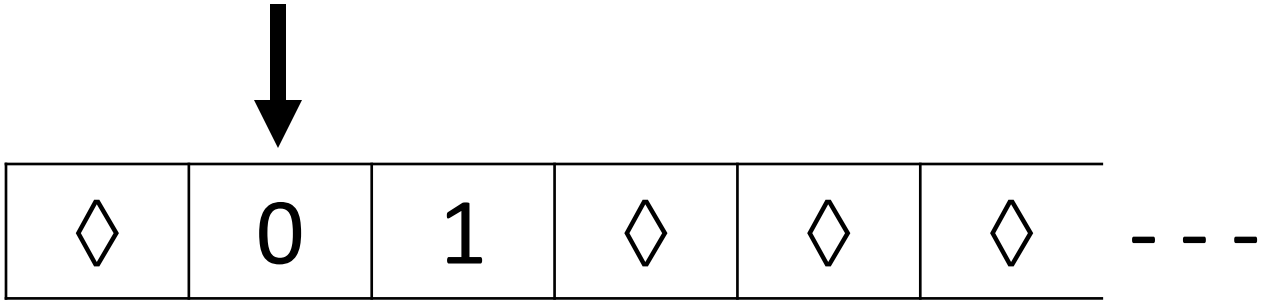
Current	Read	Action	Next
Initial	◊	**R**	Start
Start	**0**	**1**	GotZero
Start	**1**	**0**	GotOne
Start	◊	**L**	Halt
GotZero	**1**	**L**	Finish
GotOne	**0**	**R**	Carry
Carry	**1**	**0**	GotOne
Carry	**0**	**1**	GotZero
Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



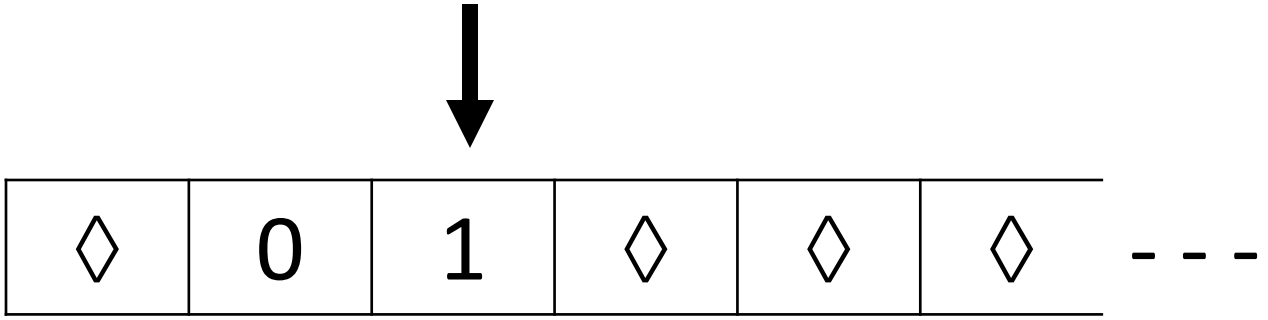
Current	Read	Action	Next
Initial	◊	**R**	Start
Start	**0**	**1**	GotZero
Start	**1**	**0**	GotOne
Start	◊	**L**	Halt
GotZero	**1**	**L**	Finish
GotOne	**0**	**R**	Carry
Carry	**1**	**0**	GotOne
Carry	**0**	**1**	GotZero
Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



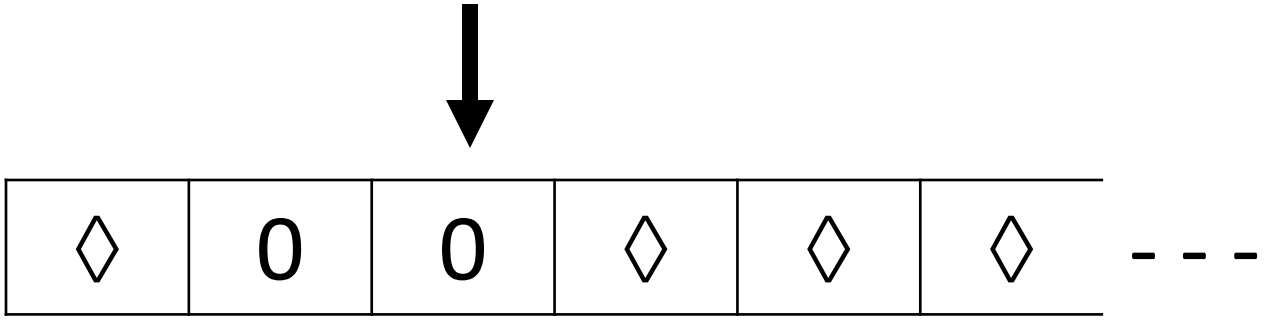
Current	Read	Action	Next
Initial	◊	**R**	Start
Start	**0**	**1**	GotZero
Start	**1**	**0**	GotOne
Start	◊	**L**	Halt
GotZero	**1**	**L**	Finish
GotOne	**0**	**R**	Carry
Carry	**1**	**0**	GotOne
Carry	**0**	**1**	GotZero
Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



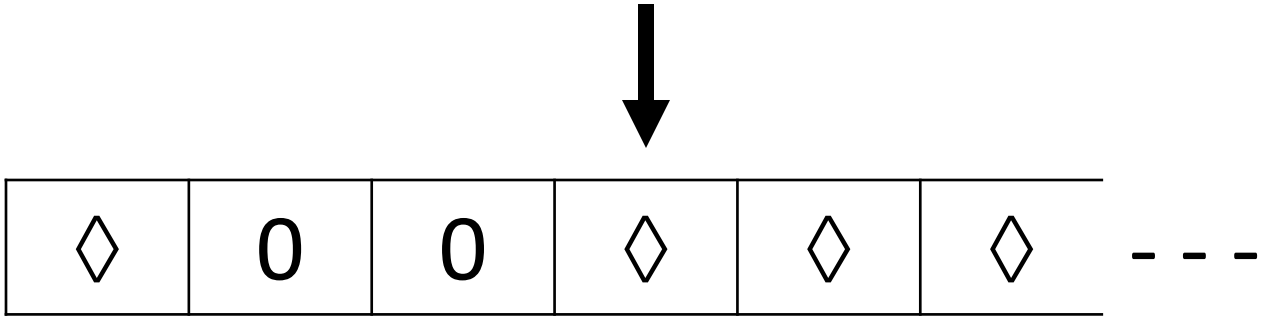
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Initial	◊	**R**	Start
Start	**0**	**1**	GotZero
Start	**1**	**0**	GotOne
Start	◊	**L**	Halt
GotZero	**1**	**L**	Finish
GotOne	**0**	**R**	Carry
Carry	**1**	**0**	GotOne
Carry	**0**	**1**	GotZero
Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



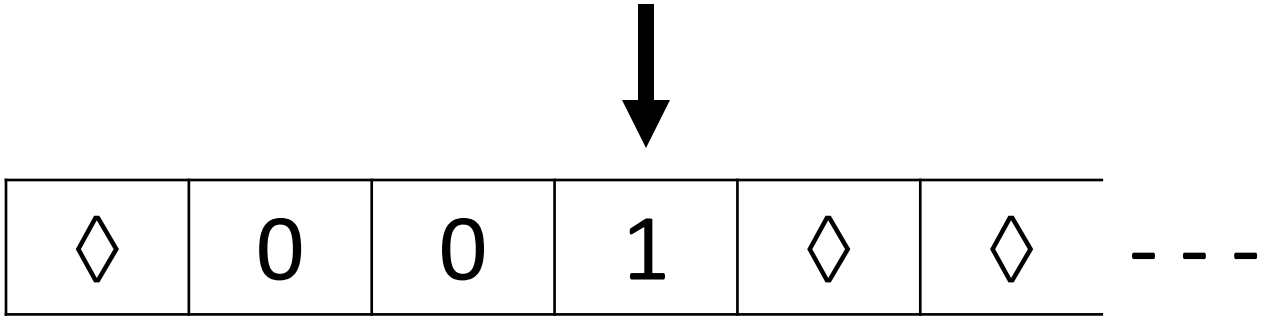
Current	Read	Action	Next
Initial	◊	**R**	Start
Start	**0**	**1**	GotZero
Start	**1**	**0**	GotOne
Start	◊	**L**	Halt
GotZero	**1**	**L**	Finish
GotOne	**0**	**R**	Carry
Carry	**1**	**0**	GotOne
Carry	**0**	**1**	GotZero
Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



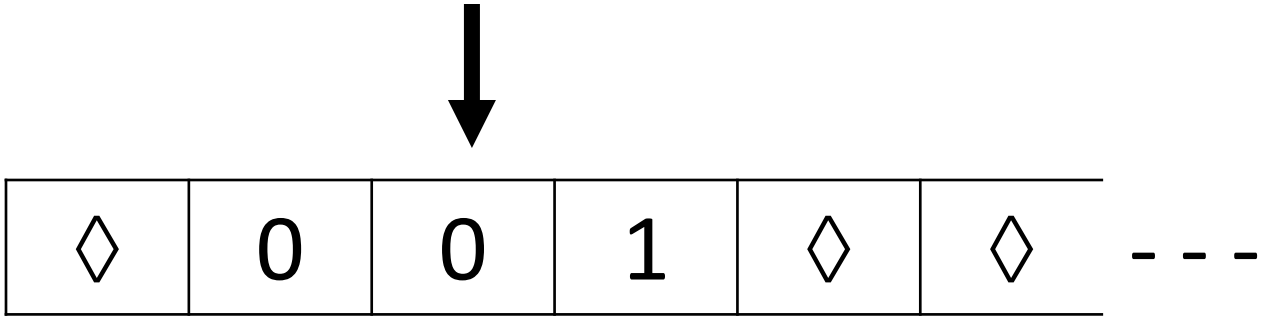
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Start	**0**	**1**	GotZero
Start	**1**	**0**	GotOne
Start	◊	**L**	Halt
GotZero	**1**	**L**	Finish
GotOne	**0**	**R**	Carry
Carry	**1**	**0**	GotOne
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Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



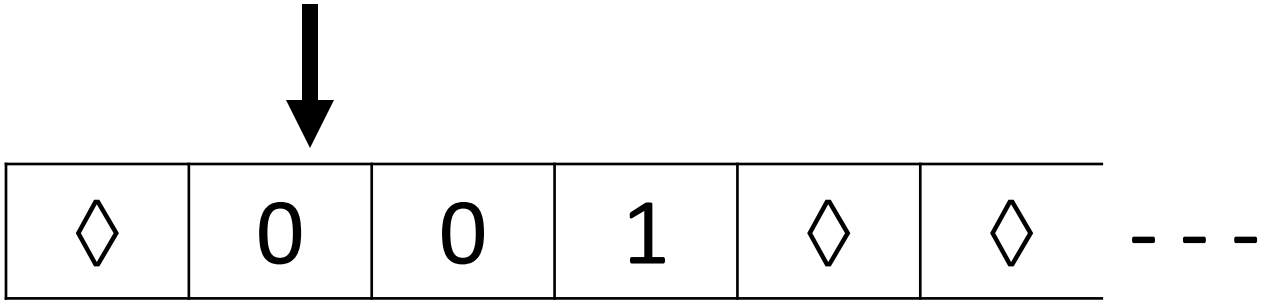
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Start	**0**	**1**	GotZero
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Carry	**1**	**0**	GotOne
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Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



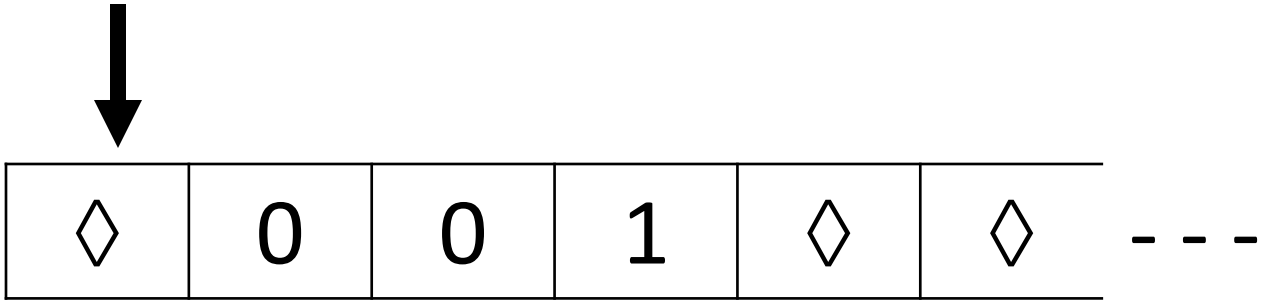
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Start	**1**	**0**	GotOne
Start	◊	**L**	Halt
GotZero	**1**	**L**	Finish
GotOne	**0**	**R**	Carry
Carry	**1**	**0**	GotOne
Carry	**0**	**1**	GotZero
Carry	◊	**1**	Finish
Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



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Initial	◊	**R**	Start
Start	**0**	**1**	GotZero
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Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



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Finish	**0**	**L**	Finish
Finish	**1**	**L**	Finish
Finish	◊	◊	Halt



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